Order Value Patterns: An Exploratory Data Analysis

Project 2 | DecodeLabs Industrial Training Kit | Dataset: 1,200 cleaned order records (Jan 2023 - Jun 2025)

1. Problem Statement

What drives order value in this e-commerce dataset, and are the largest orders genuine sales or data quality issues? Before any dashboard or predictive model is built on this data, we need to understand the shape of the numbers, confirm which extreme values can be trusted, and identify which fields actually move revenue.

2. Methodology

Starting from the Project 1 cleaned dataset (1,200 records, zero duplicates, zero missing critical fields), four numeric fields - Quantity, UnitPrice, ItemsInCart, and TotalPrice - were profiled using:

- Five-number summary (min, Q1, median, Q3, max) plus mean, standard deviation, and skewness for each field

- Outlier detection using both the IQR method (Q1 - 1.5xIQR to Q3 + 1.5xIQR) and the Z-score method (|z| > 3), to compare sensitivity on skewed vs. symmetrical fields

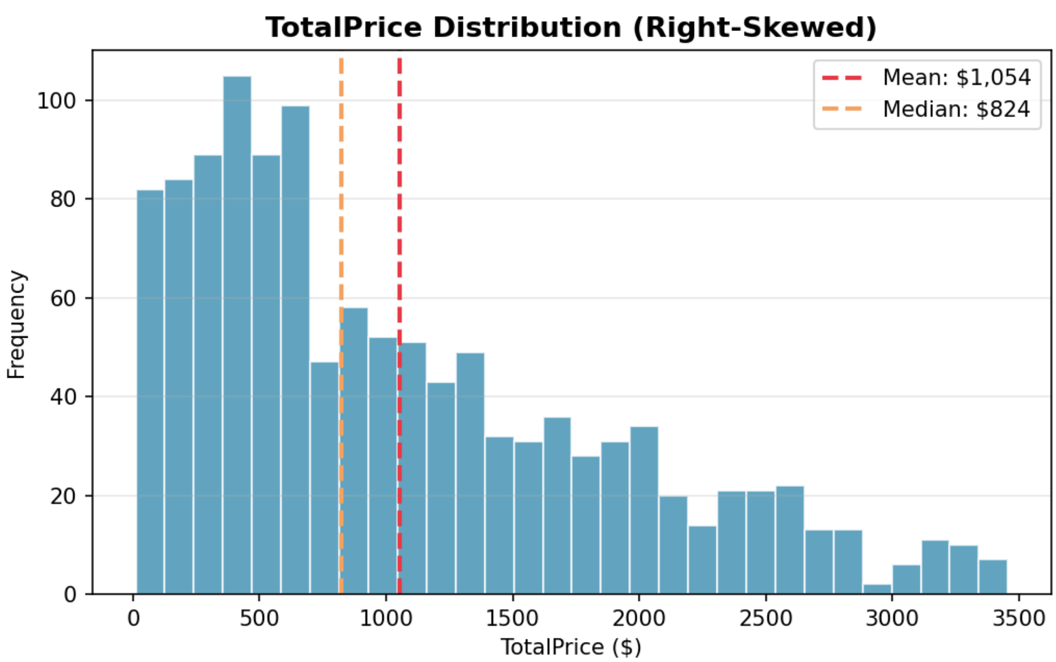
- Pearson correlation analysis across all four numeric fields to map relationships and rule out obvious confounds

- Monthly trend analysis of order count, total revenue, mean and median order value

3. Key Findings

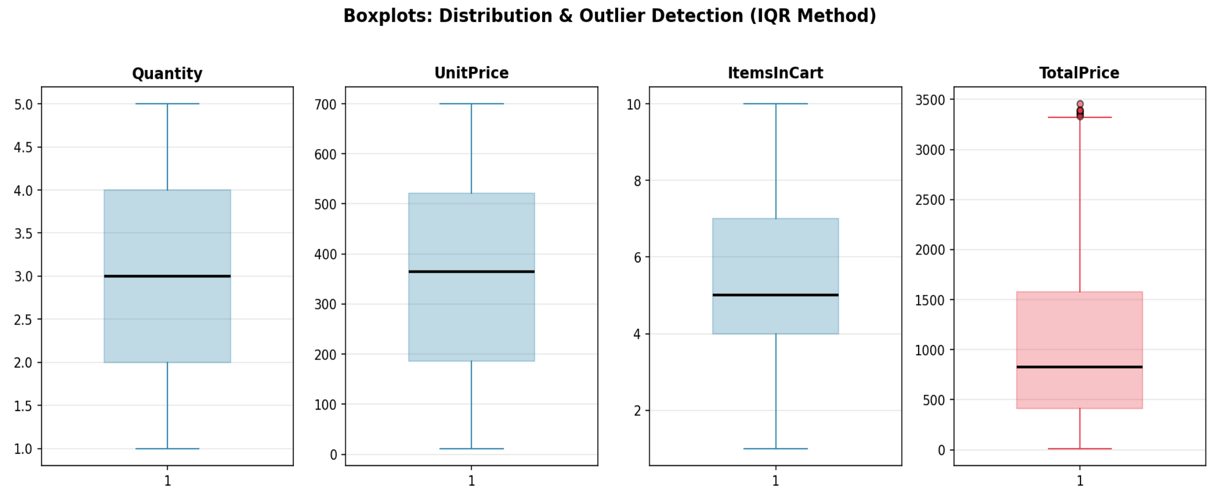
3.1 Distribution shape

Quantity, UnitPrice, and ItemsInCart are all close to symmetrical (skewness near 0). TotalPrice is right-skewed (skewness ~ 0.89) because it is the product of two independent symmetrical variables. As a result, the median ($824) is a more representative "typical order" figure than the mean ($1,054).



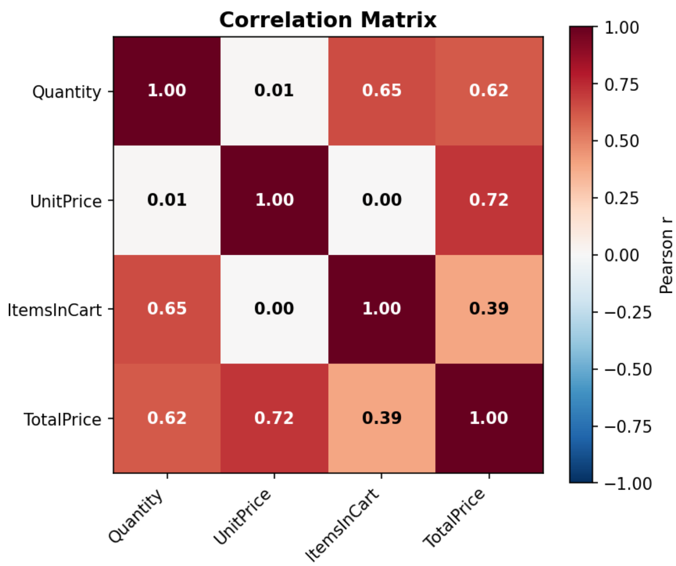
3.2 Outliers: signal, not noise

The IQR method flagged 8 orders above the upper bound on TotalPrice ($3,330.41); the Z-score method flagged zero, on any field. Every one of the 8 flagged orders has Quantity = 5 (the maximum order size) combined with a near-peak UnitPrice - they are the dataset's largest legitimate bulk orders, not data-entry errors, and were not removed. This also confirms IQR is the more trustworthy method for right-skewed money fields; Z-score assumes near-normal distributions and under-flags this kind of data.



3.3 What actually drives order value

UnitPrice correlates with TotalPrice at r = 0.72 (strong), Quantity at r = 0.62 (moderate-strong), and Quantity correlates with ItemsInCart at r = 0.65. UnitPrice and Quantity themselves show essentially no relationship (r = 0.01) - customers are not buying fewer units when unit prices are higher in this data.



3.4 Trend over time

Monthly mean order value consistently sits above the monthly median across the full Jan 2023 - Jun 2025 window, confirming the right skew is a stable structural feature of the business - driven by a recurring subset of high Quantity x high UnitPrice orders each month - rather than a one-off spike. See the Trend Analysis tab in the companion workbook for the month-by-month figures and chart.

4. Recommendations

#	Recommendation	Why
1	Report the median ($824), not the mean ($1,054), as the "typical order value" in stakeholder communications.	The mean is inflated by the right skew; the median better represents most orders.
2	Route the 8 flagged high-value orders to account management for relationship follow-up.	They are confirmed legitimate large sales, not errors - a candidate list for VIP outreach, not cleanup.
3	Prioritize unit pricing and premium product mix over pure volume discounting in revenue strategy.	UnitPrice has the stronger correlation with TotalPrice (0.72 vs 0.62 for Quantity).
4	Standardize on the IQR method for outlier screening on money fields going forward.	Z-score under-flags skewed data; IQR caught all 8 real outliers that Z-score missed.
5	Investigate cart-to-order conversion for customers with large ItemsInCart but low Quantity.	The 0.65 correlation is moderate, not perfect - a meaningful minority may be abandoning full carts.

Companion File

All calculations in this report are backed by live Excel formulas (not hardcoded values) in EDA_Analysis.xlsx: EDA Overview, Descriptive Stats, Outlier Detection, Trend Analysis, Correlation Matrix, and Visual Diagnostics tabs.